PROJECT EXPO 2026

TERRALINK

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PROBLEM STATEMENT:

During disasters and emergencies in remote or network-deprived regions, **conventional cellular communication may become unavailable or unreliable due to damaged infrastructure, network congestion, or lack of coverage.** This makes it difficult for affected people to send SOS alerts and communicate their accurate location to rescue teams.

Existing communication solutions may depend on **fixed infrastructure, continuous network availability, or costly communication systems**, which are not always practical in isolated disaster-affected areas.

Therefore, there is a need for a low-cost, portable and low-power emergency communication system that can operate without relying entirely on cellular infrastructure and can **dynamically forward SOS and GPS information through available TerraLink devices, with support from dedicated rescue relay nodes when required.**

ABSTRACT:

- **TerraLink** is a **portable emergency communication system** designed for disaster-affected people in **remote and network-deprived regions**.
- During disasters, **cellular networks may be unavailable, damaged, congested, or out of coverage**, making conventional emergency communication difficult.
- The system uses **LoRa communication and GPS** to transmit **SOS alerts along with the user's location** without depending entirely on cellular infrastructure.
- TerraLink devices can act as **temporary forwarding nodes**, allowing an SOS message to move through available nearby devices toward the rescue side.

- The system incorporates **controlled forwarding, duplicate suppression, acknowledgement, and energy-aware operation** to improve communication reliability while reducing unnecessary transmissions and battery consumption.
- **Dedicated rescue relay nodes** can be deployed by rescue/disaster-management teams when the available dynamic devices cannot provide a communication path.
- The proposed system aims to provide a **low-cost, portable, low-power, and scalable emergency communication solution** for areas where conventional communication infrastructure is unavailable or unreliable.

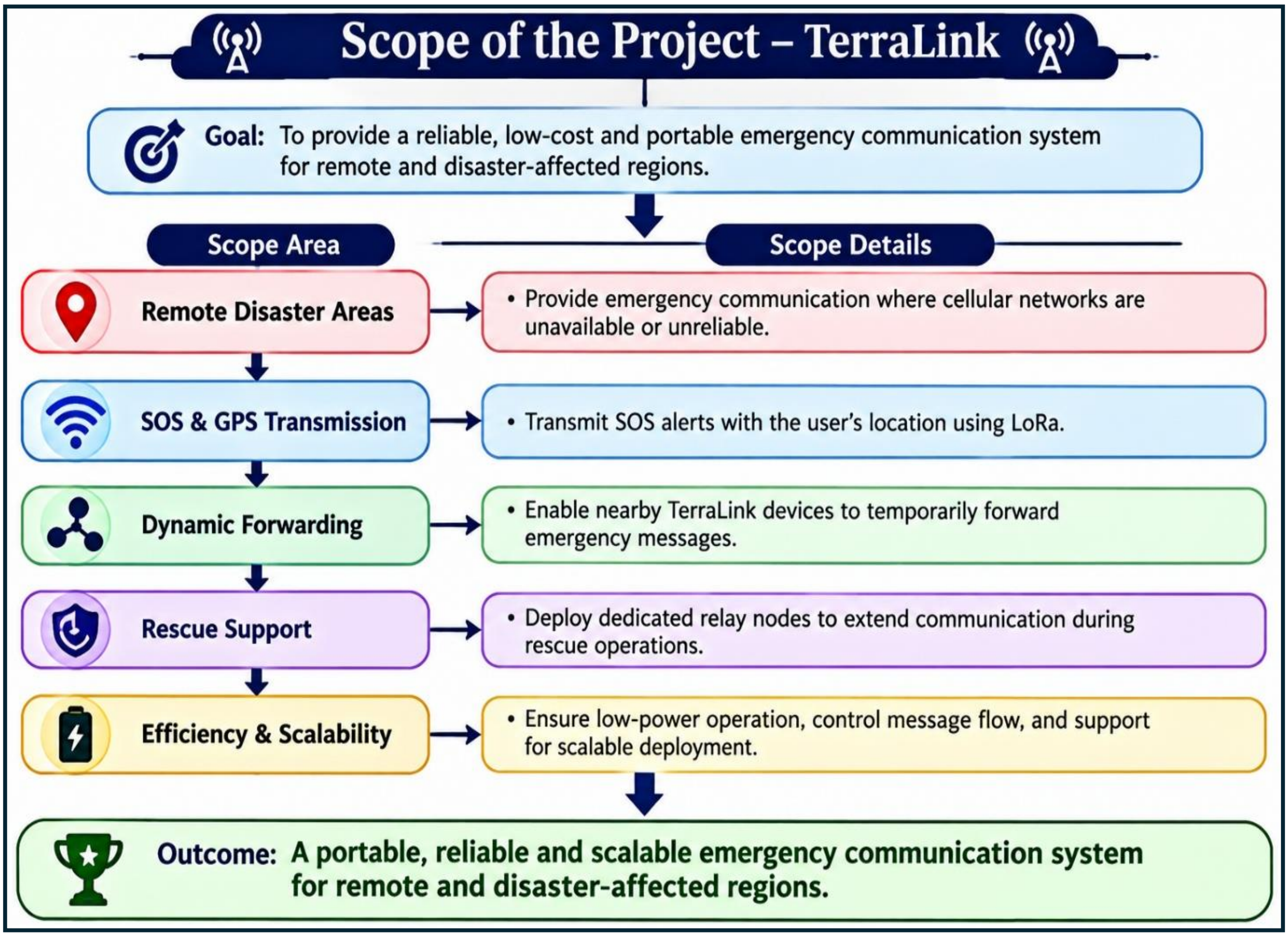
OBJECTIVES:

1. Develop a low-cost, portable emergency communication system for **remote and disaster-affected areas**.
2. **Transmit SOS alerts with GPS location** using LoRa without relying entirely on cellular infrastructure.
3. **Enable dynamic temporary forwarding** of emergency packets through available TerraLink devices.
4. **Improve communication reliability** using controlled forwarding, acknowledgement, duplicate suppression, and retransmission mechanisms.
5. **Minimize power consumption** through energy-aware communication and efficient node operation.
6. **Support rescue operations** through dedicated relay nodes when dynamic forwarding alone cannot establish a communication path.
7. **Evaluate the system** under practical conditions such as multiple SOS requests, packet collisions, limited connectivity, and varying node availability.

SCOPE OF THE PROJECT:

- **Emergency Communication:** Provide communication for people in remote and disaster-affected areas where cellular networks are unavailable or unreliable.
- **SOS & GPS:** Enable transmission of SOS alerts with the affected person's location using LoRa.
- **Dynamic Forwarding**: Allow available TerraLink devices to act as temporary forwarding nodes to extend communication beyond direct LoRa range.
- **Rescue Support**: Enable dedicated relay nodes to be deployed by rescue/disaster-management teams when additional coverage is required.

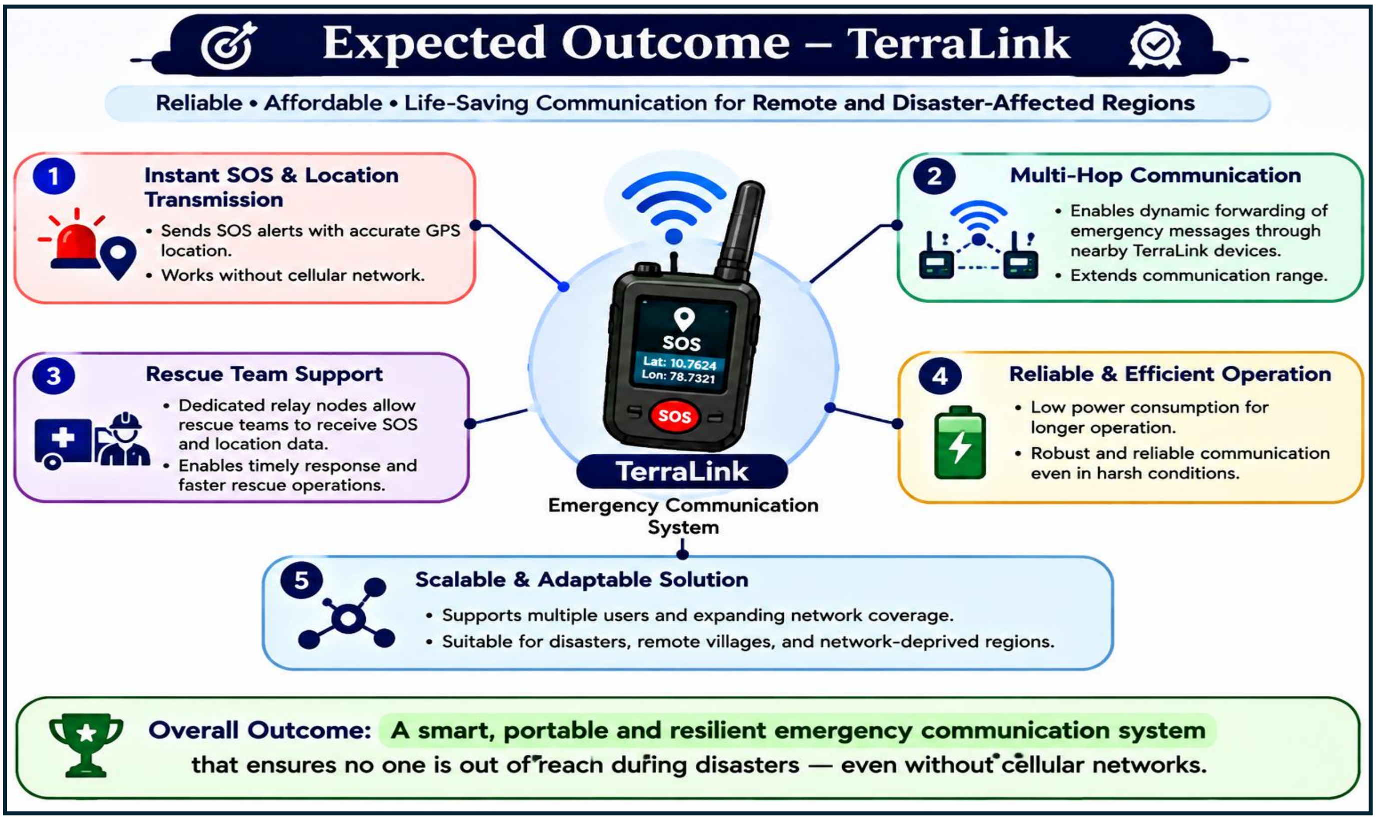
- **Reliable & Efficient Operation:** Focus on low-power operation, controlled packet forwarding, collision/flooding reduction, and scalable communication for multiple emergency situations.



EXPECTED OUTCOMES:

- **Reliable SOS Delivery:** Emergency alerts reach the rescue side with the user's **GPS location**.
- **Extended Communication:** Nearby TerraLink devices enable **dynamic multi-hop forwarding** beyond direct LoRa range.
- **Rescue Support:** Dedicated relay nodes help **rescue teams extend coverage** during emergency operations.
- **Efficient Operation: Low-power and controlled communication** reduces unnecessary battery consumption and network congestion.

- **Practical Solution:** A **low-cost, portable and scalable** communication system for remote and disaster-affected areas.



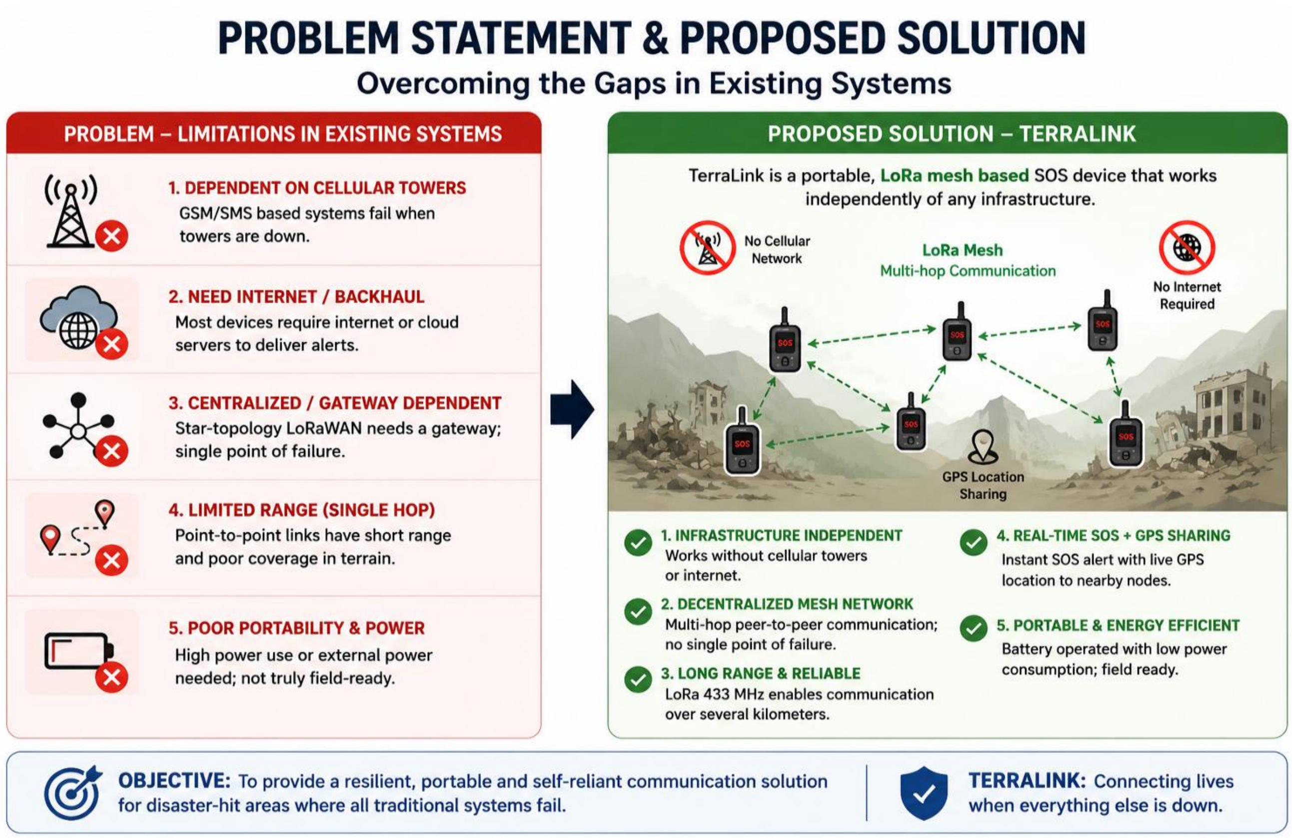
LITERATURE SURVEY

The following table surveys ten representative studies spanning LoRa-based emergency communication, wireless mesh networking, GPS-based tracking, and IoT-driven disaster-response devices, situating TerraLink within the current state of the art.

AUTHOR	CONTRIBUTION	LIMITATION
Bhatt et al.	Self-powered device relaying GPS position	Star-topology LoRaWAN architecture requiring an
Chandra et al.	Offline point-to-point communication link giving	Limited to a small number
Kumar et al.	Integrates multiple sensor streams into	Focuses on sensor fusion rather
Ferreira et al.	Enables infrastructure-less device-to-device communication between	Depends on a companion smartphone

Ochoa & Kist	Adapts the Ad-hoc On-Demand Distance	Validated mainly at small/simulated scale;
Greco et al.	Field-validated multi-hop device-to-device LoRa relay	Purpose-built for updating a fixed
Pueyo Centelles et al.	Lets citizens report predefined status	Still infrastructure-dependent on surviving gateways;
Vithayathil et al.	Proposes an SDN-managed dual-mesh smart-city	High system complexity and cost
Meyer et al.	Combines LoRa with DTN store-carry-forward	Store-carry-forward delivery introduces variable latency
Sharma & Verma	Demonstrates low-cost GPS-based location acquisition	Entirely dependent on the cellular

SOLUTION ANALYSIS:



EXISTING SOLUTION ANALYSIS:

S.No	Existing Solution	Technology Used	Working Principle	Advantages	Limitations
1	Cellular Communication	GSM, 4G, 5G, LTE	Communication through mobile towers	Low cost, high-speed data	Requires network coverage; fails during disasters
2	Satellite Communication	Iridium, Globalstar	Direct satellite-based communication	Global coverage, works in remote areas	Expensive, high power consumption
3	Satellite Emergency Messengers	Garmin, SPOT, ZOLEO	Sends SOS and GPS through satellites	Reliable emergency alerts, location tracking	High cost, subscription required
4	Walkie-Talkie Systems	VHF/UHF RF	Direct radio communication	Simple, instant, no network required	Limited range, no GPS or automatic routing
5	Personal Locator Beacons	406 MHz Beacon	Sends distress signals via satellites	Highly reliable rescue alerts	One-way communication, expensive
6	GPS Tracking Devices	GPS + GSM/LTE	Tracks location using GPS and cellular networks	Accurate tracking	Fails without cellular connectivity
7	LoRaWAN Systems	LoRa + Gateways	Long-range IoT communication through gateways	Low power, long range	Requires gateway infrastructure
8	LoRa Mesh Systems	LoRa + Mesh Protocols	Data forwarded through multiple nodes	Low power, infrastructure-free	Limited emergency-specific features

9	MANET Systems	Wi-Fi Direct, SDR, RF	Self-forming wireless networks	Self-organizing, reliable	Complex and costly
10	Drone Communication Networks	UAV + LoRa/Wi-Fi/LTE	Drones act as communication relays	Quick deployment in disasters	Limited flight time, weather dependent
11	Military Communication Systems	MANET, SDR, Encryption	Secure tactical communication	Highly reliable and secure	Very expensive and restricted
12	Research Emergency Networks	LoRa, ZigBee, IoT	Experimental disaster communication systems	Innovative, low-cost approaches	Mostly prototypes, limited deployment

GAP ANALYSIS :

Gap Analysis – TerraLink

- Cellular networks may fail or become unavailable during disasters and in remote regions, creating a communication gap.
- Fixed relay systems require permanent installation, power supply, maintenance, and infrastructure, making deployment difficult in remote areas.
- Single-hop LoRa communication has limited coverage when the affected person is beyond the direct communication range.
- Uncontrolled forwarding can lead to packet collisions, flooding, duplicate packets, and network congestion during multiple simultaneous SOS requests.
- Battery limitations affect continuous operation of portable and forwarding nodes, creating a need for energy-aware communication.
- Many emergency systems do not combine SOS, GPS location, dynamic forwarding, and rescue-side relay support in one portable solution.

REFERENCES:

1.LoRa Library for Arduino (Communication Implementation Reference)

https://github.com/sandeepmistry/arduino-LoRa

2.LoRa Mesh Networking Research Paper – IEEE Xplore https://ieeexplore.ieee.org/search/searchresult.jsp?newsearch=true&queryText=LoRa%20mesh%20network

3.LoRa Alliance – Lora WAN Technology Overview

https://lora-alliance.org/about-lora/